

Connamara Systems Introduction 2025



connamara

CELEBRATING 25 YEARS



AMERICAN FINANCIAL EXCHANGE

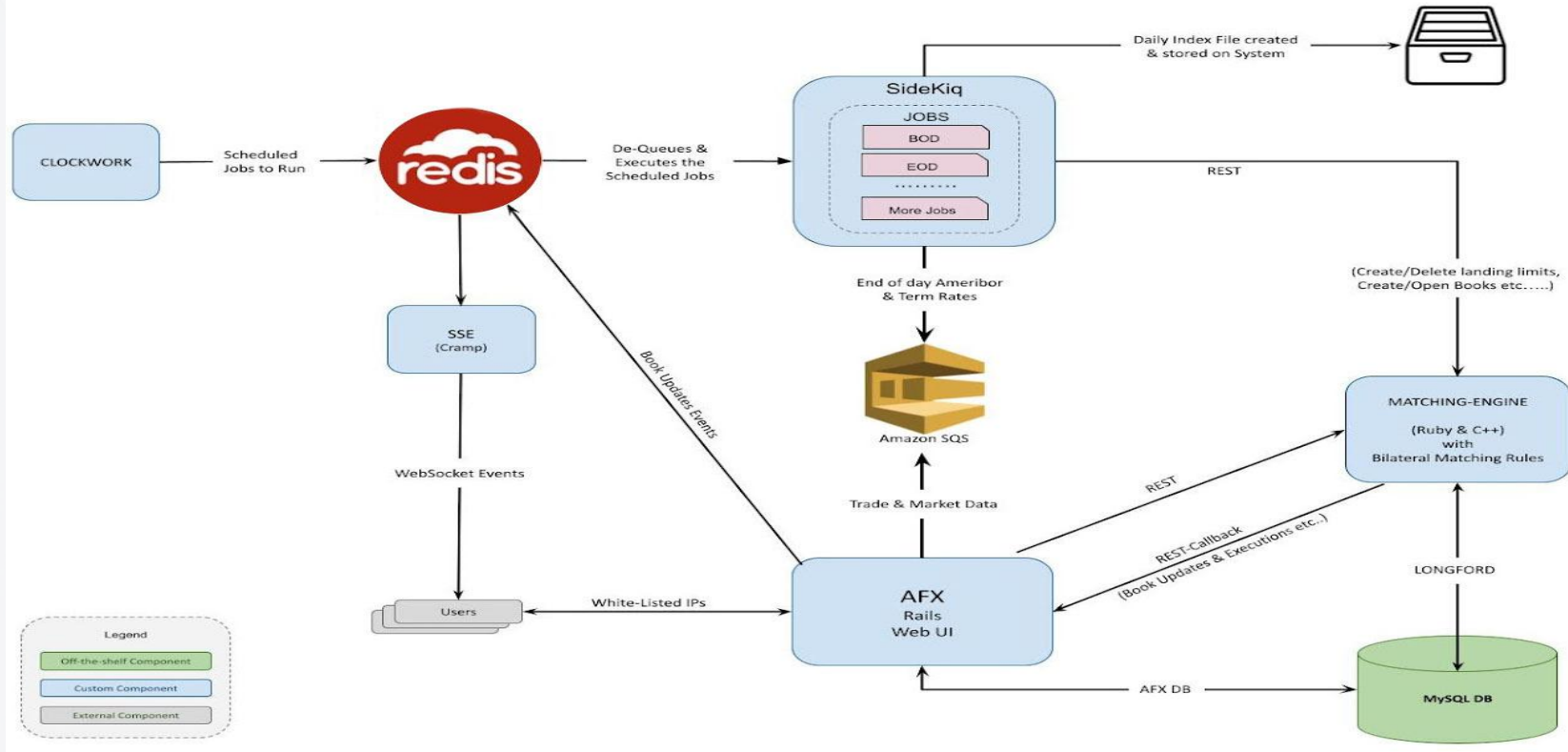
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Walkthrough System Schema

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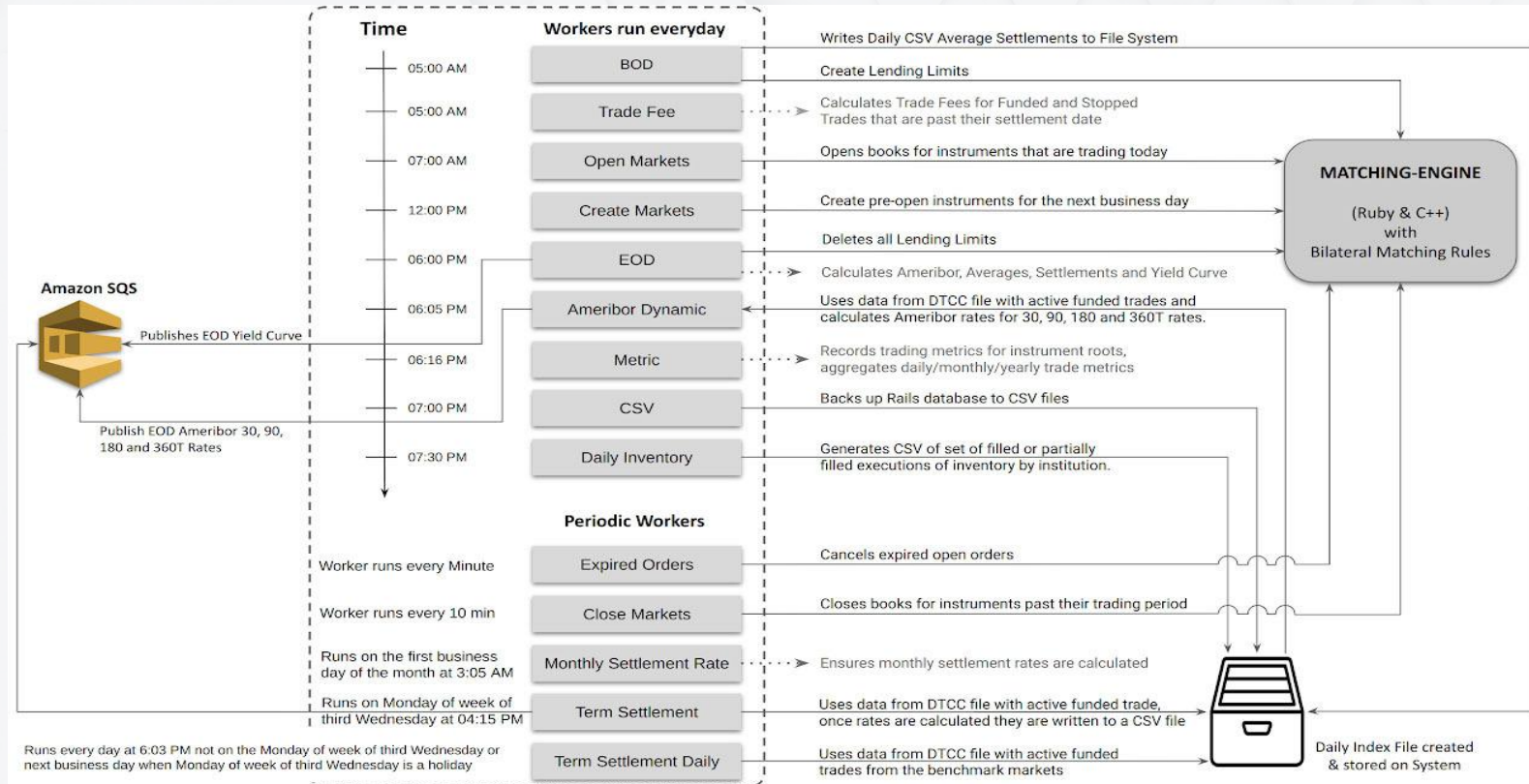
System Architecture Diagram



Components Description

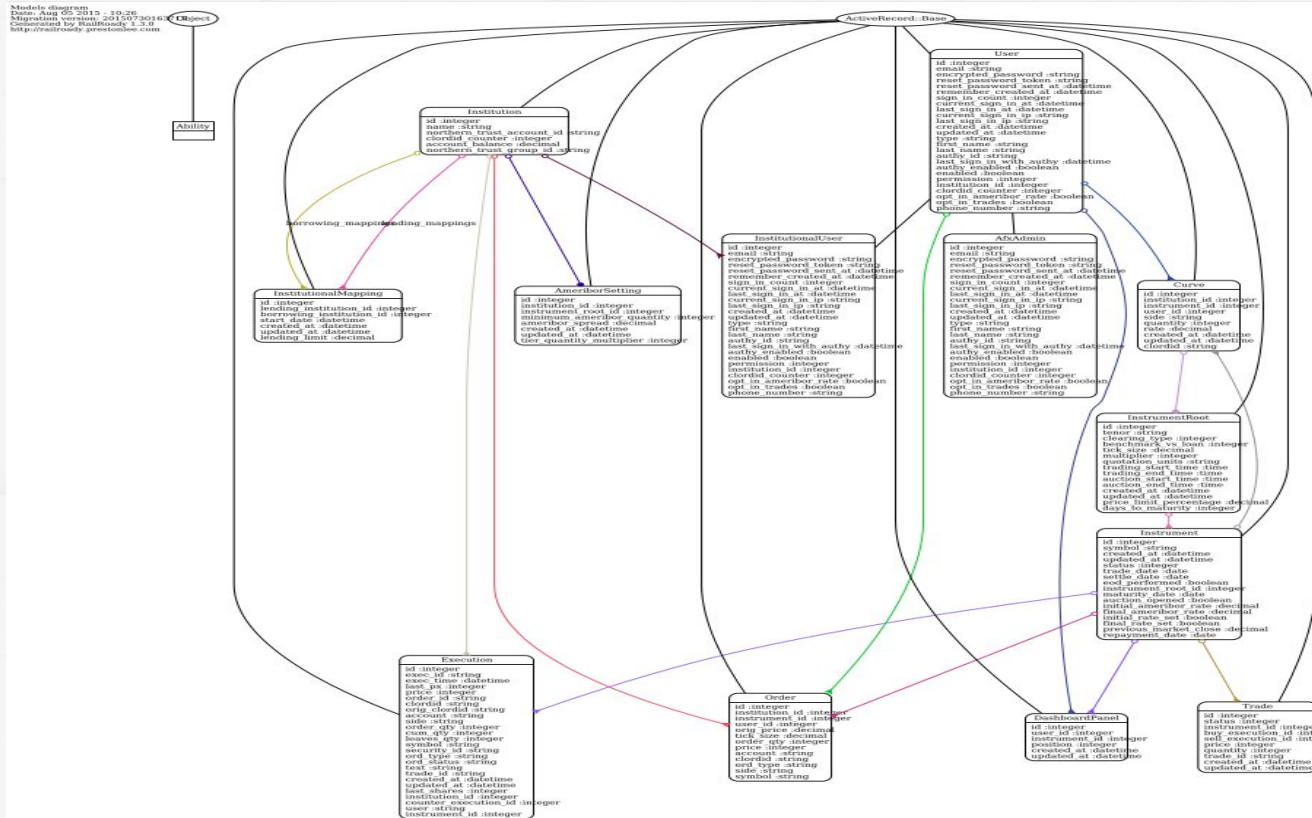
- ◆ Matching Engine - "Longford". Modified with plugin for risk, bi-lat lending logic
- ◆ Rails - In this context "rails" refers to a few different services managed by Foreman.
 - Puma - The Web-Server
 - Cramp - SSE (Server Sent Events) Server
 - Redis Heartbeat - heartbeats with redis to maintain connectivity with SSE
 - Clockwork - schedules tasks at configurable times, such as eod/bod activity
 - Sidekiq - Manages background jobs
- ◆ Redis - Used in a couple of ways
 - Used by Sidekiq as a job queue
 - Used with Cramp for real time updates to client-side Angular
- ◆ MySQL - The DB
- ◆ Amazon Simple Queue Service (SQS) - Used to queue Trade updates that can be consumed by an authorized SQS client

Scheduled Processes



DB Schema

Click on
the link to
view the
detailed
DB
Schema
[HERE](#)



Software Requirements

- Red Hat Enterprise Linux (RHEL) Version 7 - Operating System
- Ansible - Version 1.9.6
- Ansible-phantomjs - Required for automation scripts
- Apache - Web server
- MySQL - 5.6 - Database management system
- Ruby - 2.1.5 - Programming language
- Rails - 4.2.7.1 - Web application framework
- wkhtmltopdf - HTML to PDF conversion
- libselinux-python - SELinux Python bindings
- NTP - Network Time Protocol for time synchronization
- Bundle - Ruby dependency manager
- Foreman - Process manager for applications
- MySQL-python - MySQL Python connector
- Matching Engine - Internal package for matching engine
- Redis - In-memory key-value store
- postfix - Mail transfer agent
- NodeJS - JavaScript runtime
- NPM - Node.js package manager
- Git - Version control system

Supported Environments

DEMO

- This is the first environment where changes are deployed for testing.
- This is a developer centric environment where development team perform the checks
- Maintained by Connamara and is hosted in AWS

API

- This is the first environments where changes are deployed for testing.
- This is mainly for the development team to test but also allows AFX to test changes before they are moved to the next environment (CERT)

CERT

- More Production like Environment
- Primarily for clients (AFX) to test.
- Development team is not to touch this environment
- Development team accesses only when required and permitted by AFX
- Changes found working in this environment are then considered ready for Production

PROD

- The Production environment

Release Process & Schedule

Release Process

- ◆ Strictly on-need basis when updates / new features are to be released
- ◆ Changes are performed in a release-candidate branch seeded from master and named using the next version number, Ex: “2-37-release-candidate”
- ◆ Changes from Release Candidate are deployed first to DEMO for testing
- ◆ Upon successful testing are deployed to API and CERT next
- ◆ Upon further successful testing the release candidate is considered ready for PROD
- ◆ The “x-xx-release-candidate” branch is merged in master
- ◆ Main Regression Builder is passed on Jenkins
- ◆ Production Release Tag is created in Git from master
- ◆ Tag is then deployed to PROD server

Release Schedule

- ◆ Production releases are performed on Weekends (likely Saturday morning CST)

Build and Deployment Model

Build

- ◆ Jenkins is used as the Build tool
- ◆ Jenkins runs the Regression builder
- ◆ Regression builder contains all the Acceptance (Integration) Tests and Unit Tests
- ◆ Build is considered ready for deployment only when Regression Builder passes

Build and Deployment Model

Deployment

Done in 2 Steps

◆ Provisioning

- Uses Ansible
- Ansible is used to configure and provision the servers as per the Environment being deployed to

◆ Actual Deployment

- Capistrano Ruby package is used to deploy the application. (<https://capistranorb.com/>)
- Capistrano scripts bundle the Rails application, and deploy to appropriate servers
- Upon deployment services are restarted

Although the Provisioning can be performed individually using Ansible, but triggering a deployment using Capistrano will trigger the provisioning of the boxes too to make sure no configuration is broken on the boxes before the actual deployment is done

Change Tracking Tool

- ◆ GitHub is used for Issues
- ◆ Every change request is created as an issue which is then estimated and worked upon

Change Process

- ◆ Issues are created in Git
- ◆ Tagged as part of a Release
- ◆ Release candidate with appropriate next version branch is created from master. Ex: 2-37-release-candidate
- ◆ Jenkins Builder for release candidate branch is created
- ◆ Developers PR to release-candidate branch which is constantly scrutinized with builder checks upon every PR merge

QA Process

Automated Tests

- All new feature or changes are covered under automated test as much as possible
 - Unit Tests: Over 1500 unit tests
 - Acceptance Tests (cucumber): Over 1000 acceptance tests to check the system as a whole

Manual Process

- System is mainly covered with Automated tests and manual testing is done as an extra layer of testing that focuses on basic system running and testing of changes performed.
 - Manual testing process include
 - New Feature / Changes manually Tested at Connamara
 - New Feature / Changes manually Tested at AFX

Test Documents

- ◆ Informal notes of scenarios / areas tested manually are updated to AFX via the email

Technical Design Documents

- AFX Admin Documentation
- AFX Processes
- Accessing the AFX MySQL Database
- Application Log Checkup
- Managing IP Tables with Ansible
- MySQL Master-Master Replication
- MySQL Replication Failover Document

List of Projects/Actions in the Queue

- ◆ Site-to-Site VPN setup

Outline of Team Involved in 2024

- ◆ **Project Manager / Tech Lead**
 - Harvinder Vasdev
- ◆ **Developer**
 - Harvinder Vasdev
 - Nitish Bisht
- ◆ **DevOps**
 - Anshul Kumar
 - Sidharath Kapoor
- ◆ **QA**
 - Dharmendra Sharma
- ◆ **Manager**
 - Rakesh Sharma

Level of Day-to-Day Involvement

- Verify AFX Benchmark Rates
- BODProcess
- Check the Status of OpenMarket Worker
- Check the Status of CreateMarket Worker
- Check for any errors in logs

**Thank you for
your time and
attention!**

**Happy to take
any questions.**